

DSA Test - Easy Level

Instructions

- Time: 45 minutes
 - Total Questions: 5
 - No external references allowed
-

Question 1 - Pattern (Easy)

Write a program to print the following **right-angled triangle** pattern for `n = 5` :

```
*
* *
* * *
* * * *
* * * * *
```

Input: An integer `n`

Output: The above left-aligned triangle pattern

Question 2 - Pattern (Easy)

Write a program to print an **inverted right-angled triangle** for `n = 5` :

```
* * * * *
* * * *
* * *
* *
*
```

Input: An integer `n`

Output: The inverted triangle pattern

Question 3 - Hashing (Easy)

Given an array of integers, **count the frequency** of each element and print the elements with their counts.

Example:

Input: [1, 2, 2, 3, 3, 3, 4]

Output: 1→1, 2→2, 3→3, 4→1

Constraints:

- $1 \leq \text{arr.length} \leq 10^5$
 - $0 \leq \text{arr}[i] \leq 10^6$
-

Question 4 - Array (Easy)

Given an array of integers, find the **second largest** element without sorting.

Example:

Input: [12, 35, 1, 10, 34, 1]

Output: 34

Constraints:

- $2 \leq \text{arr.length} \leq 10^5$
 - Array will have at least two distinct elements
-

Question 5 - Matrix (Easy)

Given an $m \times n$ matrix, find the sum of each row and each column.

Example:

Input:

```
1 2 3
4 5 6
7 8 9
```

Output:

Row sums: 6, 15, 24

Column sums: 12, 15, 18

Constraints:

- $1 \leq m, n \leq 100$